

Probabilities and Bets

The mathematics of the ice hockey betting model for the Qminers Quant Hackathon 2025

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1. Overview

Every day the agent receives the results of the previous day's games and the bookmaker's odds for upcoming games. For each game played today it must decide whether to stake money on the home win H , the away win A or the draw D , and how much. The model works in three stages:

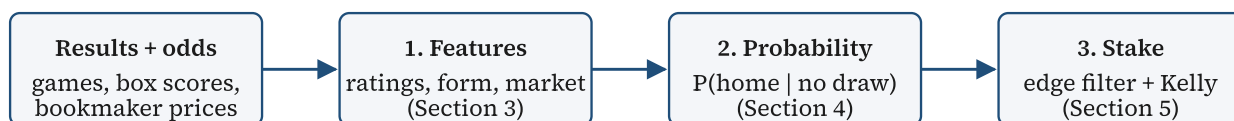


Figure 1. The daily pipeline, run inside every call of `Model.place_bets`.

The central design choice is that the bookmaker's own price is the starting point. The model never predicts outcomes from scratch: it estimates a *correction* to the market's probability, and bets only when that correction is large enough to overcome the bookmaker's margin.

Notation

Symbol	Meaning
o_H, o_A, o_D	Decimal odds for home win, away win, draw ($o_D = 0$ when a draw is impossible)
$\sigma(z) = 1/(1 + e^{-z})$	Logistic function; $\text{logit}(p) = \ln \frac{p}{1-p}$ is its inverse
i, j	Home team, away team
g_i, g_j	Goals scored by home and away team
B	Bankroll at the start of the day

2. What the bookmaker's odds say

A decimal odd o_k pays o_k per unit staked if outcome k happens, so a fair price would be $o_k = 1/P(k)$. The inverse odds $\pi_k = 1/o_k$ are therefore the bookmaker's implied probabilities, but they do not sum to one:

$$R = \sum_k \pi_k = \frac{1}{o_H} + \frac{1}{o_A} + \frac{1[o_D > 1]}{o_D} > 1.$$

The excess $R - 1$ is the bookmaker's margin (the *overround*). In this data it averages $R \approx 1.173$ from 2005 onward, a 17% margin. Normalising removes it and gives the **market probabilities**

$$q_k = \frac{\pi_k}{R}, \quad q_H + q_A + q_D = 1.$$

Why the margin matters. If you bet on outcome k and the market probability is exactly right, $P(k) = q_k$, the expected return per unit staked is

$$q_k o_k - 1 = \frac{\pi_k}{R} \cdot \frac{1}{\pi_k} - 1 = \frac{1}{R} - 1 \approx -14.7\%.$$

This holds for *every* outcome regardless of the odds, and it matches the flat-betting results in the data (between -7% and -26% across all odds ranges). To make money the model must believe an outcome is substantially more likely than the market says.

Draws only happen until the 2001 season; from 2002 onward the bookmaker offers no draw odds and every game has a winner. The model therefore splits the problem into the draw probability, taken from the market, and the **conditional home-win probability**

$$q = P_{\text{market}}(H \mid \text{no draw}) = \frac{q_H}{q_H + q_A} = \frac{1/o_H}{1/o_H + 1/o_A}, \quad m = \text{logit}(q).$$

Note that R cancels in q . The market log-odds m is the most important feature in the model.

3. Features

All team state is updated online, one finished game at a time, in date order. Before a game updates any state, its feature vector is recorded together with its result. A training row therefore contains only information that was available when the game could have been bet on. Games played on the same day are all recorded first and only then applied, so they cannot leak into one another.

3.1 Elo rating

Each team has a rating r , starting at 1500. The expected score of the home team includes a home advantage of $h_{\text{Elo}} = 60$ points:

$$E_{ij} = \frac{1}{1 + 10^{-(r_i + h_{\text{Elo}} - r_j)/400}} = \sigma\left(\frac{\ln 10}{400}(r_i + h_{\text{Elo}} - r_j)\right).$$

After the game, with actual score $S = 1$ (home win), 0 (away win) or $\frac{1}{2}$ (draw) and goal difference $d = g_i - g_j$, both ratings move by the same amount in opposite directions:

$$\Delta = K \cdot M(d) \cdot (S - E_{ij}), \quad M(d) = \begin{cases} \ln(1 + |d|) & d \neq 0 \\ 1 & d = 0 \end{cases} \quad r_i \leftarrow r_i + \Delta, \quad r_j \leftarrow r_j - \Delta$$

with $K = 8$. The factor $M(d)$ makes big wins count more, with diminishing returns. At the start of each season ratings regress toward the mean, $r \leftarrow 1500 + 0.9(r - 1500)$, because rosters change in the off-season. The feature is $\text{logit}(E_{ij})$, which by the second form above is simply the scaled rating difference.

3.2 Poisson goal ratings

Each team has an attack strength a and a defence strength δ . Goals are modelled as Poisson with expected values

$$\lambda_i = \exp(\mu + \eta + a_i - \delta_j), \quad \lambda_j = \exp(\mu + a_j - \delta_i),$$

where μ is the league scoring level (initially $\ln 3$) and η the home advantage (initially 0.1). For a Poisson observation g with mean $\lambda = e^\theta$, the log-likelihood $g\theta - e^\theta$ has gradient $\partial\ell/\partial\theta = g - \lambda$. After each game the model takes one stochastic gradient step with learning rate $\alpha = 0.02$, using the errors $\varepsilon_i = g_i - \lambda_i$ and $\varepsilon_j = g_j - \lambda_j$:

$$\begin{aligned} a_i &\leftarrow a_i + \alpha \varepsilon_i, & \delta_j &\leftarrow \delta_j - \alpha \varepsilon_i, & a_j &\leftarrow a_j + \alpha \varepsilon_j, & \delta_i &\leftarrow \delta_i - \alpha \varepsilon_j, \\ \mu &\leftarrow \mu + 0.1 \alpha \frac{\varepsilon_i + \varepsilon_j}{2}, & \eta &\leftarrow \eta + 0.1 \alpha \frac{\varepsilon_i - \varepsilon_j}{2}. \end{aligned}$$

The league-wide parameters learn ten times more slowly than the team parameters. The feature is the expected goal difference $\lambda_i - \lambda_j$.

3.3 Team form

For each team the model keeps exponentially weighted moving averages of nine per-game statistics x : goals for and against, shots on goal for and against, win (1, 0 or $\frac{1}{2}$), power-play goals for and against, minor penalties, and goalie save percentage. Each average is kept at two speeds:

$$\bar{x} \leftarrow (1 - \beta) \bar{x} + \beta x, \quad \beta_{\text{fast}} = 0.15, \beta_{\text{slow}} = 0.04.$$

The effective memory is about $1/\beta$ games: roughly 7 for the fast average and 25 for the slow one. The first observation initialises the average, and fast averages are reset at the start of each season. Two statistics are derived rather than read directly:

$$\text{shots}_i = \text{saves}_j + g_i - \text{shootout goals}_i, \quad \text{save\%}_i = \frac{\text{saves}_i}{\text{saves}_i + g_j}.$$

The first rebuilds shots on goal, which are only recorded from 2009 onward, for every season. The features are the home-minus-away differences $\bar{x}_i - \bar{x}_j$ of all nine statistics at both speeds (18 features); a team with no history contributes a difference of 0.

3.4 Schedule and market features

- **Rest:** days since the team's last game, capped at 10 (and 10 for a team's first game).
- **Back-to-back:** $1[\text{rest} \leq 1]$.
- **Season win rate:** $\frac{W + 2}{W + L + 4}$, i.e. the posterior mean under a Beta(2, 2) prior, so that early-season records are shrunk toward 50%.
- **Overround:** R from Section 2.

In total each game is described by 28 features: m , $\text{logit}(E_{ij})$, $\lambda_i - \lambda_j$, R , three schedule features per team and the 18 form differences.

4. The probability model

The target is $y = 1[\text{home win}]$, learned only from decided games, so the model estimates $p = P(H \mid \text{no draw})$. It is refit on the last 8 seasons once at least 1000 decided games are available, and again after every 150 new games.

4.1 Default: boosting from the market

The default variant (`xgb_market`) models the log-odds as the market's log-odds plus a learned correction:

$$p(x) = \sigma(m + F(x)), \quad F(x) = \nu \sum_{t=1}^{200} f_t(x),$$

where each f_t is a regression tree of depth 1 (a single split on one feature) and $\nu = 0.02$ is the learning rate. Passing m as the *base margin* to XGBoost means boosting starts from the market's prediction: with $F \equiv 0$ the model is the market. The trees are fit by gradient boosting on the log-loss

$$\mathcal{L} = - \sum_n \left[y_n \ln p(x_n) + (1 - y_n) \ln (1 - p(x_n)) \right],$$

regularised by an L2 penalty of 5 on leaf values, a minimum of 20 games per leaf, and subsampling 80% of rows and columns per tree.

Interpretation. Because every tree has depth 1, F is a sum of step functions of single features, $F(x) = \sum_k F_k(x_k)$, with no interactions. The model is therefore an additive correction to the market log-odds, close in spirit to a generalised additive model. Starting from the market and allowing only small additive corrections is what keeps it from re-learning what the bookmaker already knows and from overfitting.

4.2 Alternative: logistic regression

The other variants use L2-regularised logistic regression on standardised features z , $p = \sigma(\beta_0 + \beta^\top z)$, with inverse regularisation strength C . They were compared with the default in walk-forward validation (Section 6).

4.3 Optional adjustments

Two further options exist for experiments; both are switched off in the default model. Training rows can be weighted by season age, $w = \gamma^{S_{\text{last}} - S}$, and predictions can be shrunk toward the market in log-odds space, $z' = m + s(z - m)$ with $0 \leq s \leq 1$.

4.4 From conditional to outcome probabilities

The draw probability is taken from the market, $p_D = q_D$ (zero whenever no draw odds are offered), and combined with the model's conditional probability:

$$P(H) = p(1 - p_D), \quad P(A) = (1 - p)(1 - p_D).$$

The model never bets on draws.

5. Making bets

5.1 Which outcome

The expected return per unit staked on outcome k is the **edge**

$$e_k = P(k) o_k - 1.$$

For each game played today the model takes the outcome with the larger expected payout, $k^* = \arg \max_{k \in \{H, A\}} P(k) o_k$, and bets only if $e_{k^*} \geq 0.05$. Combining this with Section 2, a bet requires

$$\frac{P(k)}{q_k} \geq 1.05 R \approx 1.23 \quad (R = 1.173) :$$

the model has to rate the outcome about 23% more likely than the market does. This is why it bets rarely.

5.2 How much: the Kelly criterion

Staking a fraction f of the bankroll on an outcome with probability P and odds o multiplies the bankroll by $1 + f(o - 1)$ on a win and by $1 - f$ on a loss. The Kelly criterion chooses f to maximise the expected logarithmic growth

$$G(f) = P \ln(1 + f(o - 1)) + (1 - P) \ln(1 - f).$$

Setting $G'(f) = \frac{P(o - 1)}{1 + f(o - 1)} - \frac{1 - P}{1 - f} = 0$ gives

$$f^* = \frac{P o - 1}{o - 1} = \frac{e}{o - 1}.$$

Full Kelly is optimal only if P is known exactly. Here P is an estimate whose errors are of the same size as the edge, and over-betting is much more costly than under-betting (G falls steeply beyond f^* and is negative beyond about $2f^*$). The model therefore stakes a quarter of Kelly, with caps:

$$\text{stake} = \min\left(B \cdot \min\left(\frac{1}{4}f^*, 0.05\right), \text{max bet}, \text{remaining daily budget}\right).$$

The daily budget starts at $0.2 B$ and decreases with each bet placed that day. A stake below the minimum bet is dropped. Bets are only placed on the game day itself, when the ratings include every earlier result.

5.3 Worked example

A game with $o_H = 1.45$, $o_A = 2.30$, no draw odds, bankroll $B = 1000$, bet limits 5 to 100.

Step	Computation	Value
Implied probabilities	$\pi_H = 1/1.45$, $\pi_A = 1/2.30$	0.690, 0.435
Overround	$R = \pi_H + \pi_A$	1.124
Market probability	$q = \pi_H/R$, $m = \text{logit}(q)$	0.613, 0.461
Model output (assumed)	$F(x) = 0.585$, $p = \sigma(m + F)$	0.740
Edges	$e_H = 0.740 \cdot 1.45 - 1$, $e_A = 0.260 \cdot 2.30 - 1$	+0.073, -0.402
Bet?	$e_H \geq 0.05$	yes, on H
Kelly fraction	$f^* = 0.073/0.45$	0.162
Stake	$1000 \cdot \min(0.162/4, 0.05)$	40.56

The market gives the home team 61.3%; the model believes 74.0%, a ratio of 1.21 against the required $1.05 R = 1.18$ for this game. Even a correction this large leads to a stake of only about 4% of the bankroll.

6. How the model is evaluated

A probability model can only make money if it is better than the market, so the main metric is the log-loss gain over the market on decided games, evaluated walk-forward: for each test season the model is fit only on earlier seasons.

$$\text{gain} = \mathcal{L}_{\text{market}} - \mathcal{L}_{\text{model}}, \quad \mathcal{L} = -\frac{1}{N} \sum_n \left[y_n \ln p_n + (1 - y_n) \ln(1 - p_n) \right].$$

Variant	Gain, all seasons	Gain, 2002+	Bankroll, all seasons	Bankroll, 2007–2011
xgb_market (default)	+0.0023	+0.0005	1160.55	987.83
logreg_compact	+0.0001	+0.0005	924.00	894.57
xgb_all	+0.0012	−0.0016	1017.95	941.59
logreg	−0.0008	−0.0005	1070.74	1000.00
logreg_all	−0.0015	−0.0044	556.99	691.23

Bankrolls start at 1000 and come from the full daily simulation (`src/evaluate.py`). The 2007–2011 column gives the model all earlier games as history and bets only from 2007 on, as the submission system did with the hidden seasons.

The default model's log-loss is only about 0.0005 better than the market's in recent seasons. Section 5.1 shows why that is not enough: a bet needs the model to rate an outcome about 23% above the market, and an improvement of this size almost never produces such a gap. Correctly, the model bets very rarely in recent seasons and roughly breaks even. Further experiments (training only on the no-draw era, shrinking toward the market, calibration checks and feature selection, in `src/experiments.py`) did not find a larger edge.

Summary of constants

Component	Parameters
Elo	$K = 8$, home advantage 60, season carry-over 0.9, base 1500
Poisson ratings	$\alpha = 0.02$, $\mu_0 = \ln 3$, $\eta_0 = 0.1$
Form	$\beta_{\text{fast}} = 0.15$, $\beta_{\text{slow}} = 0.04$
Training	last 8 seasons, at least 1000 games, refit every 150 games
XGBoost	200 depth-1 trees, $\nu = 0.02$, L2 penalty 5, min. 20 per leaf, 80% subsampling
Staking	min. edge 0.05, $\frac{1}{4}$ Kelly, max. 5% per game, max. 20% per day